

Defensive Drone Policy Matrix

Updated view: autonomy is separated from shared-state regimes, and the scenario knobs are called out explicitly.

Targeting Policies

Assignment and Control Regimes

	Info shared across defenders			No info shared
	Corridor Defenders stay inside their assigned angular sector.	Collaboration Corridor-first, then idle neighbors help overloads.	Global Any defender can engage any live attacker.	Autonomy NEW No shared state between defenders after launch.
Optimized <i>Weighted matching</i> Ranks by intercept time and deadline risk.	Optimized Corridor Scores threats inside the owned sector with a near-capture lock.	Optimized Collab Runs corridor matching first, then lets idle neighbors assist overloads.	Optimized Global Uses the same scoring with a fully shared matching pool.	Optimized Autonomy Each defender applies the score locally from its own observations.
Nearest <i>Distance greedy</i> Chooses the closest live attacker first.	Nearest Corridor Greedy nearest-target pursuit within the owned sector.	Nearest Collab Corridor-first nearest rule, then idle neighbors help nearby overloads.	Nearest Global Greedy nearest-target pursuit across the full attacker set.	Nearest Autonomy Each defender independently chases its nearest perceived attacker.
Earliest Deadline <i>Urgency greedy</i> Chooses the shortest time-to-target attacker.	Deadline Corridor Targets the most urgent attacker inside the owned sector.	Deadline Collab Applies urgency inside corridor, then assists the most urgent leftovers.	Deadline Global Urgency-first targeting across the full shared attacker set.	Deadline Autonomy Each defender independently selects the most urgent perceived threat.

Scenario framing and policy taxonomy are separated so experiment assumptions do not get conflated with the control rules.

Update Highlight Adds Autonomy as a fourth regime. Separates shared and no-shared-state control.	Measurement Primary metric: attackers that reach the target (missed or breached).	Scenario Presets 300 attackers; 10-100 defenders (step 10); start range about 5 miles.	Variable Knobs Attacker speed, defender speed, attack load, and distance from target.
--	---	--	---